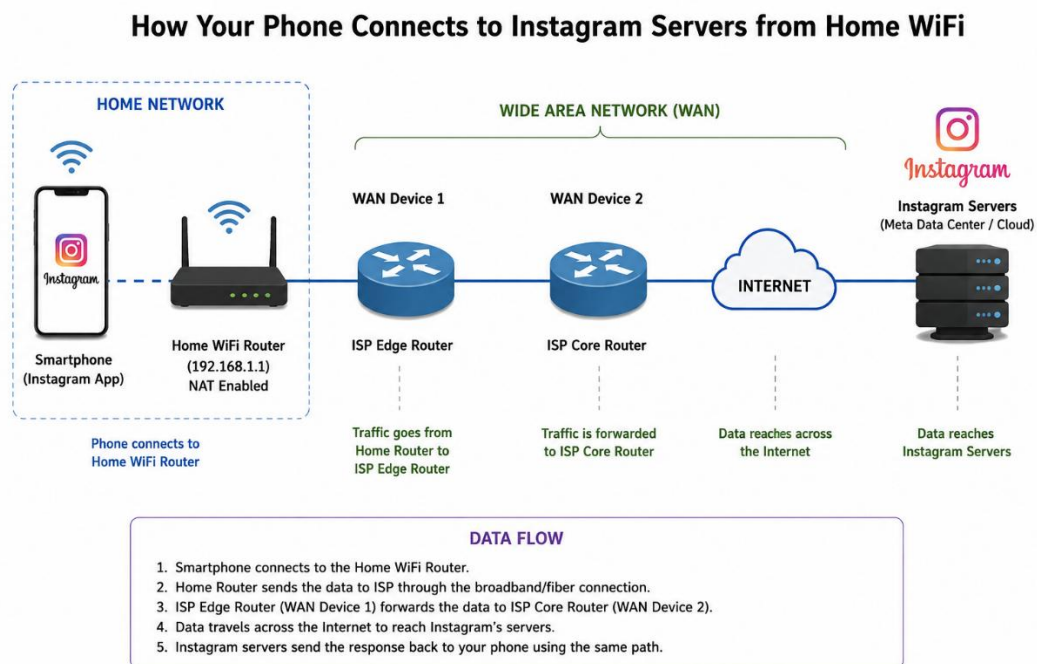


1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology	Speed	Cost	Real-Life Use
MPLS (Multiprotocol Label Switching)	High	High	Used by large companies like Zomato offices , banks, and corporate branches for secure communication.
Leased Line	Very High (Dedicated Connection)	Very High	Used by IRCTC ticketing systems , data centers, hospitals, and corporate offices where reliable internet is required.
Broadband Internet	Medium to High	Low	Used in homes , schools, small businesses, and Wi-Fi connections.

**3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Hint: Focus on how the VPN creates a private tunnel over the public WAN.**

A **VPN (Virtual Private Network)** creates a **secure and encrypted tunnel** over the public **WAN (Internet)**.

Example:

Suppose you are using **public Wi-Fi in a coffee shop** to access **Spotify or YouTube**.

Without a VPN:

- Your data travels over the public internet.
- Hackers on the same Wi-Fi may be able to intercept your data.

With a VPN:

- Your device connects to a VPN server.
- The VPN encrypts your data and creates a **private tunnel** over the public WAN.
- Even if someone intercepts the traffic, they cannot read the encrypted data.
- This keeps your browsing secure and protects your personal information.

Benefits of VPN:

- Encrypts internet traffic.
- Protects data on public Wi-Fi.
- Hides your IP address.
- Improves privacy and security.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Best WAN Technology:

MPLS (Multiprotocol Label Switching)

Reasons:

1. **Secure Communication:** MPLS provides a private and secure network for transferring business data between warehouses.
2. **High Performance:** It offers low latency and reliable connectivity, making inventory updates and order processing fast and efficient.